

Section 5.2 Independence and Dimension

Exercise 5.2.6

Use Theorem 5.2.3 to determine if the following sets of vectors are a basis of the indicated space.

b. $\{(1, 1, -1), (1, -1, 1), (0, 0, 1)\}$ in $\mathbb{R}^3$

e. $\{(2, 1, -1, 3), (1, 1, 0, 2), (0, 1, 0, -3), (-1, 2, 3, 1)\}$ in $\mathbb{R}^4$

Exercise 5.2.7

In each case show that the statement is true or give an example showing that it is false.

a. If $\{x, y\}$ is independent, then $\{x, y, x + y\}$ is independent.

c. If $\{y, z\}$ is dependent, then $\{x, y, z\}$ is dependent for any x .

f. If $ax + by + cz = 0$, then $\{x, y, z\}$ is independent.

k. Every set containing a spanning set is again a spanning set.

Exercise 5.2.10

If $\{x_1, x_2, x_3, x_4, x_5, x_6\}$ is an independent set of vectors, show that the subset $\{x_2, x_3, x_5\}$ is also independent.

Exercise 5.2.20

Let U and W denote subspaces of $\mathbb{R}^n$, and assume that $U \subseteq W$. If $\dim W = 1$, show that either $U = \{0\}$ or $U = W$.

Section 5.4 Rank of a Matrix

Exercise 5.4.1

In each case find bases for the row and column spaces of A and determine the rank of A .

a. $A = \begin{bmatrix} 2 & -4 & 6 & 8 \\ 2 & -1 & 3 & 2 \\ 4 & -5 & 9 & 10 \\ 0 & -1 & 1 & 2 \end{bmatrix}$

d. $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ -3 & -6 & 3 & -2 \end{bmatrix}$

Exercise 5.4.2

In each case find a basis of the subspace U .

a. $U = \text{span}\{(1, -1, 0, 3), (2, 1, 5, 1), (4, -2, 5, 7)\}$

d. $U = \text{span}\left\{\begin{bmatrix} 1 \\ 5 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ -8 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ -10 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix}\right\}$